

Assignment-2 Session14

Question 1: Given the following pseudo-code for a function that calculates delivery charges based on order amount, write white-box test cases to achieve 100% Statement Coverage.

Pseudo Code

```
function calculateDeliveryCharge(amount):  
    if amount >= 500:  
        return 0  
    else:  
        return 50
```

Answer

Objective

Cover every executable statement in the code at least once.

Test Cases

TC ID	Input (Amount)	Expected Result	Statements Covered
TC-001	₹600	Return 0	if block
TC-002	₹400	Return 50	else block

Coverage Achieved

- ✓ if statement executed
- ✓ return 0 executed
- ✓ else statement executed
- ✓ return 50 executed

Statement Coverage = 100%

Question 2: Write white-box test cases for the following pseudo-code to achieve 100% Branch Coverage.

Pseudo Code

```
function getDiscount(userType, total):  
    if userType == "student":  
        if total > 1000:
```

```

        return 20
    else:
        return 10
else:
    return 5

```

Answer

Objective

Execute every True and False branch.

TC ID	User Type	Total Amount	Expected Result	Branch Covered
TC-001	student	1200	Return 20	Outer IF = True, Inner IF = True
TC-002	student	800	Return 10	Outer IF = True, Inner IF = False
TC-003	regular	1500	Return 5	Outer IF = False

Coverage Achieved

✓ Outer IF (True)

✓ Outer IF (False)

✓ Inner IF (True)

✓ Inner IF (False)

Branch Coverage = 100%

Question 3: For the following pseudo-code, write test cases to achieve Condition Coverage.

Pseudo Code

```

function isPaymentAllowed(method, amount):
    if (method == "UPI" or method == "Card") and amount <= 10000:
        return true
    else:
        return false

```

Answer

Conditions

C1 = method == "UPI"

C2 = method == "Card"

C3 = amount <= 10000

Each condition must evaluate to **True** and **False** at least once.

TC ID	Method	Amount	Expected Result	C1	C2	C3
TC-001	UPI	5000	True	T	F	T
TC-002	Card	8000	True	F	T	T
TC-003	Cash	5000	False	F	F	T
TC-004	UPI	15000	False	T	F	F

Coverage Achieved

C1 → True & False ✓

C2 → True & False ✓

C3 → True & False ✓

Condition Coverage = 100%

Question 4: Given the pseudo-code below for a simple login check, write white-box test cases to cover all unique execution paths (Path Coverage).

Pseudo Code

```
function login(username, password):  
  if username == "admin":  
    if password == "pass123":  
      return "Login Success"  
    else:  
      return "Wrong Password"  
  else:  
    return "User Not Found"
```

Answer

Possible Execution Paths

Path 1

Username = admin
Password = pass123

Output: Login Success

Path 2

Username = admin
Password = wrong password

Output: Wrong Password

Path 3

Username $\neq$ admin

Output: User Not Found

Test Cases

TC ID	Username	Password	Expected Result	Path Covered
TC-001	admin	pass123	Login Success	Path 1
TC-002	admin	123456	Wrong Password	Path 2
TC-003	saksham	pass123	User Not Found	Path 3

Coverage Achieved

✓ Path 1 executed

✓ Path 2 executed

✓ Path 3 executed

Path Coverage = 100%

Summary

Coverage Technique	Objective	Test Cases Used
Statement Coverage	Execute every statement at least once	2
Branch Coverage	Execute every True and False branch	3
Condition Coverage	Make every condition evaluate to True and False	4
Path Coverage	Execute every unique execution path	3

Conclusion

This assignment demonstrates four important **White Box Testing** techniques:

Statement Coverage ensures every executable statement is executed at least once.

Branch Coverage verifies all decision branches (True and False) are tested.

Condition Coverage checks that each individual condition evaluates to both True and False.

Path Coverage validates every possible execution path through the program.

Using these techniques helps improve code quality, identify hidden defects, and achieve better test coverage during software development.